Indications and Timing of Surgeries in Congenital Heart Diseases

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Introduction

Congenital heart diseases (CHDs) are common birth defects worldwide, with a prevalence of 8–12/1000 live births. It accounts for third commonest cause of death in neonates.

Timely intervention in CHD is crucial for improving outcomes. In this paper, we present a summary of guidelines for the management of common CHDs in children.

CHDs are traditionally classified as follows based on pulmonary blood flow (PBF):

Acyanotic CHD

- ☑ Left to right shunts (Increased PBF) – ASD, VSD, PDA, APW, PAPVC
- ☑ Obstructive/regurgitant lesions (Normal PBF) – AS, PS, CoA, MR, AR

Cyanotic CHD

- ☑ Decrease PBF—TOF physiology
- ☑ Increase PBF—TGA, truncus arteriosus, TAPVC

ACYANOTIC CHD

Left to Right Shunt Lesions (Table 1)

TABLE 1: Clinical quantification of left to right shunt.

	Small shunt	***Moderate shunt***	***Large shunt***
Precordium	Normal	Hyperdynamic	Hyperdynamic
Heart size	Normal size heart	Cardiomegaly	Cardiomegaly
Second heart sound (P2)	Normal P2	Normal or loud P2 (P2<A2)	Loud P2 (P2>A2)
Pulmonary arterial hypertension (PAH)	No PAH	No PAH or sub-systemic PA pressures	Systemic PA pressures

Left to Right Shunt Lesions

Atrial Septal Defect (ASD)

It is of 4 types: Ostium secundum (75%), ostium primum (15–20%), sinus venosus (5–10%) and coronary sinus (<1%).

Ventricular Septal Defect (VSD)

Classified based on location in interventricular septum into perimembranous, muscular, inlet and outlet types.

Patent Ductus Arteriosus (PDA)

- ☑ In term neonates, although small PDA may close up to 3 months of age, the chances of spontaneous closure of large PDA are rare as it occurs due to a structural abnormality of the ductal smooth muscle.
- ☑ On the other hand, PDA in preterm neonates closes spontaneously in 35–75% cases before the first birthday. Hemodynamically significant PDAs are managed medically with intravenous or oral drugs like indomethacin, ibuprofen and paracetamol. There is no role for prophylactic drug therapy. Surgical ligation is indicated when two courses of medical therapy fail or the drugs are contraindicated.

TABLE 2: Summary of timing of closure and follow-up of left-to-right shunts.

	ASD	***VSD***	***PDA***
Indications and ideal age	L to R shunt with RV volume overload *Asymptomatic:* 2–4 years *Symptomatic in infancy:* early closure *Those presenting late after ideal age:* Elective closure, if operable	*Small VSD:* Only if associated complications like endocarditis or cusp prolapse with AR, RVOT obstruction *Moderate VSD:* Asymptomatic (2–5 yrs); symptomatic and controlled with medications (1–2 yrs) *Large VSD:* Poor growth/ uncontrolled CHF (as early as possible); controlled CHF (6 months) Any VSD with cusp prolapse and AR–immediate surgery	*Small PDA:* 12–18 months *Moderate PDA:* No CHF (6 months – 1 year); CHF (early, by 3 months) *Large PDA:* Early by 3 months
Contra-indication	Irreversible PVD (pulmonary vascular disease)	Irreversible PVD	Irreversible PVD, Silent PDA (PDA with no murmur)
Methods	*Device closure:* For OS-ASD with good rims *Surgical repair:* All other types of ASD, OS-ASD with deficient rim <5 mm	*Device closure:* Mid and anterior muscular VSDs, postoperative residual VSD, PM-VSD at least 4 mm away from aortic valve *Contraindicated* in associated AR, conduction abnormalities *Surgery:* Patch closure (standard method); *Staged PA banding* –only in multiple VSDs, inaccessible VSDs and children with contraindications for CPB	*Device closure:* Preferred in child with weight >4 kg to 6 kg (institutional policy) *Surgical ligation:* Child with weight < 4–6 kg, PDA with unusual shape, progressively enlarging or symptomatic ductal aneurysm, endarteritis
Follow-up	*After device closure:* Antiplatelets for 6 months; ECG and ECHO at discharge, 1 month, 6 months, 1 year and then 3–5 yearly *After surgical closure:* ECHO in first year, follow-up later if any complaints - IE prophylaxis for 6 months after ASD closure	*After device closure:* Antiplatelets for 6 months; ECG and ECHO at discharge, 1 month, 6 months, 1 year, annually till 5 years and then 3–5 yearly *After surgical closure:* ECG, ECHO in first year, follow-up later if any complaints - IE prophylaxis for 6 months after VSD closure	*After surgical closure:* ECG, ECHO in 1st year, follow-up later if any complaints - IE prophylaxis for 6 months after PDA closure

Left to right Shunt Lesions

Atrioventricular Septal Defect (AVSD)/Endocardial Cushion Defect

A complete AVSD has a large septal defect with an atrial component (ostium primum defect), a large ventricular component (inlet septal defect), a common atrioventricular (AV) valve ring, and a common AV valve. On the other hand, a partial AVSD has an ostium primum ASD with separate AV valve rings, and usually cleft of the anterior leaflet of the AV valve.

Aortopulmonary Window (APW)/ Aortopulmonary Septal Defect

APW is a defect between the ascending aorta and the main pulmonary artery. It is rare and comprises around 0.1% of all CHDs.
Indications for closure:

- ☑ *Uncontrolled CHF:* Immediate repair of the defect
- ☑ *Controlled CHF:* Elective repair by 3 months of age

Surgical patch repair is the treatment of choice in most cases, while transcatheter device closure is feasible in certain cases of intermediate type APW (central defect with adequate rims).

Partial Anomalous Pulmonary Venous Connection (PAPVC)

PAPVC is a CHD in which one or more, but not all, of the pulmonary veins are connected to systemic venous tributaries. Surgical repair is indicated in the presence of a large left to right shunt. If the child has a single anomalous pulmonary vein with an intact atrial septum and a very small shunt, it may be left unoperated.

- ☑ *Coarctation of aorta (CoA)*
- ☑ *Aortic stenosis:* AS can be valvular (80–85%), subvalvular (15%) or supravalvular (least common).
- ☑ *Pulmonary stenosis:* It can be valvular (80–90%), supravalvular or subvalvular.

TABLE 3: Summary of timing of repair and follow-up of obstructive lesions.

	CoA	***AS***	***PS***
Indications, ideal age and methods	*Indications:* CoA gradient ≥20 mm Hg (peak) with diastolic tailing LV dysfunction LVH Upper limb HTN >50% narrowing relative to aortic diameter at diaphragm *Ideal age:* ☑ LV dysfunction, CHF, severe upper limb HTN: immediate ☑ Normal LV function, no CHF, mild upper limb HTN: 3–6 months ☑ Normal LV function, no CHF/HTN: 1–2 years *Methods:* *Surgical:* Neonatal presentation, infants with native CoA *Balloon angioplasty:* Critically ill neonates, infants with recoarctation, children <25 kg with native CoA/ recoarctation *Catheter based stenting:* Children >25 kg with native CoA/recoarctation	*Valvular AS* *Immediate surgery or balloon dilation:* Neonates with critical AS, LV dysfunction *Elective balloon dilation:* ☑ Peak gradient >64 mm Hg, mean gradient >40 mm Hg, ☑ Symptomatic AS or ST changes in ECG even with lower gradients *Subvalvular AS* *Surgery:* ☑ Peak gradient ≥50 mm Hg ☑ If peak gradient is less than <50 mm Hg, if associated AR of more than mild severity, LV dysfunction or plan to involve in competitive sports *Supravalvular AS* *Surgery:* ☑ Peak gradient ≥64 mm Hg/ mean gradient ≥50 mm Hg, ☑ Lower gradients but symptomatic , LV dysfunction, severe LVH, evidence of MI	*Valvular PS* *Immediate surgery or balloon dilation:* Neonates with critical PS, RV dysfunction *Elective balloon dilation:* Peak gradient >64 mm Hg, mild RV hypoplasia causing hypoxia, dysplastic pulmonary valve *Subvalvular and supravalvular* PS *Surgery:* All indications same as mentioned above for valvular PS *Peripheral PS* Focal branch and/or peripheral PA stenosis with >50% diameter narrowing, RV systolic pressure >50 mm Hg (or >50% of systemic pressure), difference in perfusion of both lungs of >20% (on lung perfusion scan), symptomatic child *Balloon dilatation ± stenting* – treatment of choice *Surgery* – when not amenable to balloon dilatation

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Obstructive Lesions (Table 3)

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	CoA	***AS***	***PS***
Follow-up	Life-long follow-up – initially annual, 2–3 yearly later BP measurement, ECHO at each visit ☑ IE prophylaxis for 6 months after intervention ☑ β blockers preferred as anti-hypertensive	Life-long follow-up with ECG, ECHO Periodic INR for those with valve replacement ☑ IE prophylaxis in those with prosthetic valve	Life-long follow-up with ECG, ECHO Periodic INR for those with valve replacement ☑ IE prophylaxis in those with prosthetic valve

Regurgitant Lesions (Table 4)

Aortic Regurgitation (AR)

Congenital AR may be due to a bicuspid aortic valve, or may occur during the course of VSD or subaortic membrane.

Mitral Regurgitation (MR)

Isolated congenital MR is rare. It may occur in association with AVSD.

Regurgitant Lesions (Table 4)

TABLE 4: Summary of timing of repair and follow-up of AVSD and regurgitant lesions.

	AVSD	***AR and MR***
Indications	*Complete AVSD:* Controlled CHF (by 3 months); uncontrolled CHF (as early as possible) *Partial AVSD:* Normal PA pressures (2–3 years), associated AV valve regurgitation (early) *Contraindication:* Irreversible PVD	*AR* *Symptomatic* with moderate to severe AR *Asymptomatic* with severe AR if LVEF <50% or LVESD z-score >4 *MR* *Symptomatic* with moderate to severe MR *Asymptomatic* with severe MR if LVEF <60%, LVESD z-score >3 or PA systolic pressure >50 mm Hg
Method/types	☑ *Complete surgical repair* ☑ *PA banding:* Reserved for complex cases and in patients with contraindications for CPB	☑ *Valve repair:* Preferred ☑ *Valve replacement:* Bioprosthetic valve or prosthetic
Follow-up	*After surgical closure:* ECG and ECHO Annually till 10 years, followed by 2–3 yearly ☑ IE prophylaxis for 6 months after AVSD closure	Life-long follow-up ECG, ECHO every 1–2 yearly Oral anticoagulation for 3 months (bioprosthetic) and life-long (metallic) ☑ IE prophylaxis in those with prosthetic valve

Miscellaneous Lesions

Anomalous Origin of the Left Coronary Artery from the Pulmonary Artery (ALCAPA)

ALCAPA is characterized by retrograde blood flow from the right coronary artery to the left coronary artery through inter-coronary collaterals and then to the pulmonary artery resulting in myocardial steal and left ventricular infarction. ALCAPA will present in infancy usually at 2–4 months of age with clinical features of congestive cardiac failure. It is often misdiagnosed as viral myocarditis/dilated cardiomyopathy. All children with ALCAPA require immediate surgical correction to establish a two-coronary system.

CYANOTIC CHD

Congenital Cyanotic Heart disease (CCHD) can be grouped into

- ☑ *Cyanotic CHD with decreased pulmonary blood flow,* e.g.: Tetralogy of Fallot, single ventricle with PS, pulmonary atresia, tricuspid atresia with PS, Ebstein anomaly
- ☑ *Cyanotic CHD with increased pulmonary blood flow,* e.g.: Transposition of great arteries, total anomalous pulmonary venous drainage, truncus arteriosus, single ventricle without PS

TABLE 5: Summary of timing of repair and follow-up of CCHD with reduced pulmonary blood flow.

	TOF/ TOF like(VSD-PS) physiology	***Single ventricle with PS***	***Ebstein anomaly of tricuspid valve***
Indications	☑ Cyanosis (Saturation <80%), ☑ Recurrent cyanotic spells *Contraindication* Hypertensive MAPCAS, ventricular dysfunction	☑ Cyanosis (Saturation <80%), ☑ Recurrent cyanotic spells *Contraindications* Severe ventricle dysfunction or mitral regurgitation	Desaturation – SpO_2 - <90%, Cardiomegaly (CTR >0.65), RV dysfunction on ECHO
Timing and methods	*<4 months with severe cyanosis/ Recurrent cyanotic spells* –Palliative modified Blalock – Taussig shunt Or RVOT/PDA stenting *4–6 months with cyanosis:* TOF repair surgery (VSD closure + RVOT resection +/– transannular patch) *Pink TOF:* 6–12 months of age (institutional policy)	*Stage 1:* Neonatal period – PDA stenting/Modified BT shunt (if neonatal severe hypoxia) *Stage 2:* At 6–12 months – bidirectional Glenn shunt *Stage 3:* At 4–6 Years – Fontan completion	*Surgical repair:* Tricuspid valve repair – 'cone repair' at 1–3 years of age *Severe forms:* Needs individualized care (single ventricle protocol)
Follow-up	☑ Life-long follow-up ☑ Clinical assessment, ECG (for QRS duration), ECHO (RV dilatation, aortic root dilatation) at discharge, 1, 3, 6 months and then yearly till adulthood and then 2–3 yearly ☑ IE prophylaxis life-long ☑ Good oral hygiene	Life-long follow-up ECG, ECHO every 1 yearly ☑ Oral anticoagulation life-long (post Fontan) ☑ Antiplatelets life-long (post Glenn shunt) ☑ IE prophylaxis life-long	Life-long follow-up ECG, ECHO every 1–2 yearly Oral anticoagulation for 3 months (repair) and life-long (metallic) ☑ IE prophylaxis in those with prosthetic valve

Tetralogy of Fallot (TOF)

Four different variants in TOF have been described with differences in anatomy, clinical presentation and surgical management; (a) Classical TOF, (b) TOF with pulmonary atresia, (c) TOF with aortopulmonary collaterals, (d) TOF with absent pulmonary valve.

Cyanotic Heart Disease with Increase Pulmonary Blood Flow

Transposition of Great Arteries (d-TGA)

There are two major subsets for TGA

- ☑ *Simple TGA:* In this subset, TGA is not associated with VSD or outflow tract obstruction
- ☑ *Complex TGA:* In this subset, TGA is associated with large VSD/PDA and/or outflow tract obstruction.

Persistent Truncus Arteriosus (TA)

Congenital cyanotic CHD in which, a single great vessel arises from heart as a common arterial trunk giving origin to aorta and pulmonary artery.

Total Anomalous Pulmonary Venous Connection (TAPVC)

- ☑ Pulmonary veins drains anomalously into right heart via different routes. There are 4 variants depending on site of drainage namely *supracardiac, cardiac, infradiaphragmatic or mixed* TAPVC
- ☑ If there is obstruction in pulmonary venous drainage, its can present immediately at birth. In symptomatic neonates, it may mimic like persistent pulmonary hypertension and hyaline membrane disease (reticular pattern on chest X-ray).

TABLE 6: Summary of timing of repair and follow-up of cyanotic heart disease with increased pulmonary blood flow.

	TGA	***Truncus arteriosus***	***TAPVC***
Indications	All patients with TGA except in those with irreversible pulmonary vascular disease presenting at later age	All patients except in irreversible pulmonary vascular disease presenting at later age	All patients diagnosed with TAPVC except in those with irreversible pulmonary vascular disease presenting at later age
Contra-indication	Irreversible PVD	Irreversible PVD	Irreversible PVD—very unlikely
Ideal age	*Simple TGA* ☑ *Asymptomatic:* Within 2–4 weeks of life ☑ *Symptomatic:* As early as possible *TGA with large VSD* At 6–8 weeks (more risk of early PVD) *TGA with CoA-* - As early as possible *Those presenting late after ideal age* Elective closure, if operable	*Uncontrolled heart failure* - As early as possible *Controlled heart failure* At 3–6 weeks (more risk of early PVD) *TA with CoA/IAA-* - As early as possible *Those presenting late after ideal age* Elective repair, if operable clinically or on cardiac catheterization	*Obstructed TAPVC* - Emergency surgery *Unobstructed TAPVC:* Electively as early as possible when diagnosed—might present late even after 1 year of age *Those presenting late after ideal age* Elective repair, if operable clinically or on cardiac catheterization
Methods	☑ *Arterial switch operation (ASO):* If LV not regressed ☑ *Two stage arterial switch operation:* Borderline LV regression presenting between 2 and 4 months of age ☑ *Atrial switch operation(Senning/ Mustard operation):* Those presenting late with regressed LV	☑ *Truncus repair – VSD closure + RV to PA conduit +/– truncal valve repair* ☑ Need for future conduit revision *surgeries to be discussed*	*Surgical TAPVC repair*
Follow-up	☑ Life-long follow-up ☑ Antiplatelets for 3 months after ASO ☑ Clinical assessment, ECG, ECHO at discharge, 1, 3, 6 months and then yearly till adulthood and then 2–3 yearly ☑ IE prophylaxis for 6 months after surgery	☑ Life-long follow-up ☑ Clinical assessment, ECG, chest X-ray, ECHO at discharge, 1, 3, 6 months and then yearly till adulthood and then 2–3 yearly ☑ IE prophylaxis life-long ☑ Good oral hygiene	☑ Clinical assessment, ECG, ECHO at discharge, 1, 3, 6 months and then yearly till 5 years or later if residual lesion ☑ IE prophylaxis for 6 months if no residual defect ☑ Good oral hygiene

CCHD with increased pulmonary blood flow in situations with univentricular heart, present with heart failure and minimal cyanosis. Surgical pathway includes pulmonary artery banding at 6–8 weeks, followed by Glenn and Fontan surgery later.

Further Reading

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Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Rickettsial Disease

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